CONCEPT

System Architecture

```
┌──────────────────────────────────────────────────────────────────┐
│                    APACHE AIRFLOW (Docker)                       │
│                                                                  │
│  ┌──────────┐    ┌──────────┐    ┌──────────┐    ┌──────────┐   │
│  │  FETCH   │──▶│  CLEAN   │──▶│  SAVE    │──▶│  PRINT   │   │
│  │ WEATHER  │    │TRANSFORM│    │  CSV     │    │ SUMMARY  │   │
│  └──────────┘    └──────────┘    └──────────┘    └──────────┘   │
│       │                                                          │
│  Open-Meteo API                                                  │
│  (Free, No API Key)                                              │
│  latitude & longitude                                            │
└──────────────────────────────────────────────────────────────────┘
```

Pipeline Flow — Step by Step

Task 1 — fetch_weather

```
Open-Meteo API ──▶ HTTP GET Request
                        |
                        ▼
                   Parse JSON Response
                        |
                        ▼
                   Extract Fields:
                   - city name
                   - temperature (°C)
                   - windspeed (km/h)
                   - wind direction (°)
                   - weather code
                   - timestamp
                        |
                        ▼
                   XCom Push ──▶ Next T
```

Task 2 — clean_weather

```
Raw Weather Data ──▶ Round temperature to 1 decimal
                          |
                          ▼
                     Round windspeed to 1 decimal
                          |
                          ▼
                     Translate weather code
                     to human description:
                     0  ──▶ Clear sky
                     2  ──▶ Partly cloudy
                     3  ──▶ Overcast
                     61 ──▶ Slight rain
                     95 ──▶ Thunderstorm
                          |
                          ▼
                     Clean Data ──▶ XCom Push
```

Task 3 — save_to_csv

```
Clean Weather Data ──▶ Check if file exists
                                |
                          ┌─────▼─────┐
                          |  New?     |
                          └─────┬─────┘
                       Yes ─────┴───── No
                        |               |
                 Write Header     Append Only
                        |               |
                        └───────┬───────┘
                                ▼
                        weather_data.csv ✅
```

Task 4 — print_summary

```
Clean Weather Data ──▶ Format Summary
                              |
                              ▼
                    ======== WEATHER SUMMARY ========
                      City        : London
                      Temperature : 12.0°C
                      Wind Speed  : 15.2 km/h
                      Direction   : 230°
                      Condition   : Partly cloudy
                      Timestamp   : 2026-04-05 22:00:00
                    ==================================
                              |
                              ▼
                         Print to Airflow Logs ✅
```

⚙️ Tech Stack

Layer	Technology	Purpose
Orchestration	Apache Airflow 3.1.8	Pipeline scheduling & monitoring
Containerization	Docker	Isolated runtime environment
Data Source	Open-Meteo API	Free live weather data
Storage	CSV	Lightweight data export
Language	Python 3.12	Pipeline logic
HTTP Client	Requests	API calls
Data Format	JSON	API response format

🔁 Scheduling & Automation

```
Every hour Airflow scheduler triggers the pipeline:

00:00 ──▶ Pipeline starts
00:01 ──▶ Weather data fetched from Open-Meteo API
00:02 ──▶ Data cleaned and transformed
00:03 ──▶ Results saved to CSV
00:04 ──▶ Summary printed to logs
          Pipeline complete ✅

Next run: 01:00 ──▶ repeats automatically
```

🌍 Why Open-Meteo?

```
✅ 100% Free forever
✅ No API key required
✅ No signup needed
✅ No rate limits for personal use
✅ Covers worldwide locations
✅ Updates every hour
```

Just pass latitude and longitude in the URL:

```
https://api.open-meteo.com/v1/forecast
?latitude=51.5
&longitude=-0.12
&current_weather=true
```

📁 Project Structure

```
airflow-test/
├── docker-compose.yaml        # Airflow Docker setup
├── .env                       # Environment variables
├── dags/
|   └── weather_pipeline.py    # Weather Pipeline DAG
├── logs/
|   └── weather_data.csv       # Collected weather data
└── plugins/                   # Airflow plugins
```